Reactivity impacts of fuel depletion

23 LESSON

Read: FNRP § 10.4–10.5; Eqs. (10.27)–(10.39).

Objective: Quantify the evolution of nuclides and core reactivity with burnup.

1 Review: reactivity in time

As a reactor operates (at power), its excess reactivity typically decreases because ^{235}U is consumed and parasitic fission products (FPs) are born. However, that excess reactivity must be balanced by control.

To represent these processes, define a time-varying multiplication factor as

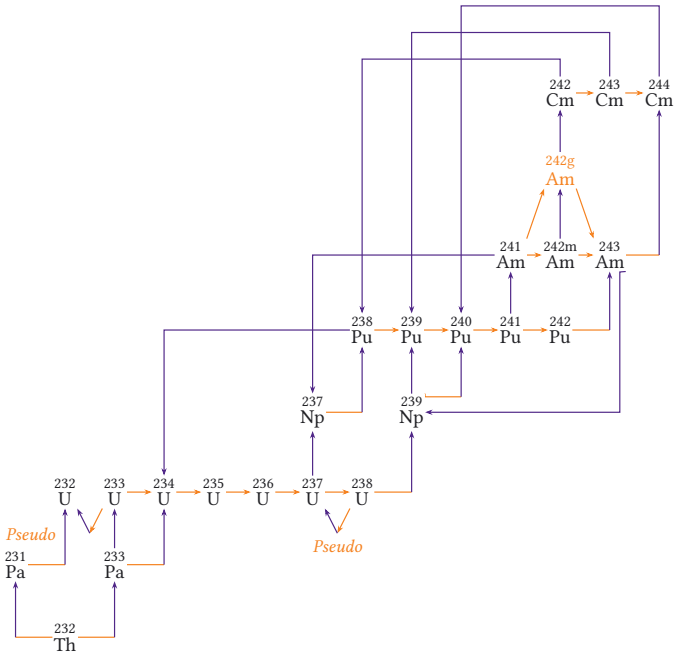
$$k(t) = \frac{\nu\bar{\Sigma}_{fT}^{F}(t)\, p\, \varepsilon P_{\mathrm{NL}}}{\bar{\Sigma}_{aT}^{F}(t) + \bar{\Sigma}_{aT}^{\mathrm{FP}}(t) + \varsigma\frac{V_M}{V_F}\left(\bar{\Sigma}_{aT}^{M} + \bar{\Sigma}_{aT}^{\mathrm{control}}(t)\right)}, \qquad \text{(FNRP 10.4*)}$$

where P_{NL} is the probability that neutrons do not leak from the system. The asterisk marks the explicit control term.

During operation, $\bar{\Sigma}_{aT}^{\mathrm{control}}(t)$ must be adjusted to maintain $k(t) = 1$. Here, adjustable control is introduced within the moderator volume, consistent with typical LWRs, whether by changing the rod position or soluble poison concentration.

2 A representative depletion chain

In theory, all nuclides connected through capture or decay should be simultaneously modeled as functions of time. A practical depletion chain for analysis of LWRs is shown below: each arrow *into* a nuclide represents production, while an *outward* arrow represents destruction.



Adapted from the IAEA WIMS-D chain. Orange: neutron-induced production; purple: decay. Mixed (orange+purple) paths collapse immediately decaying intermediates. The superscripts m and g refer to metastable and ground states; orange ^{242g}Am uses reduced-concentration bookkeeping for its branch to ^{242}Cm. *Pseudo* denotes a fictitious fission product that is introduced to account for non-negligible contributions from $(n, 2n)$ interactions. See the IAEA WIMS-D documentation for details.

3 Further simplifications

Because each fissile nuclide has unique fission-product distributions, tracking the entire isotopic inventory remains quite complex! A much more drastic simplification is presented in FNRP based on these assumptions:

1. Only ^{235}U, ^{238}U, and ^{239}Pu are important.

2. Ignore their decay.

3. ^{239}Pu is produced *directly* and *immediately* from ^{238}U(n,γ) reactions.

4. The amount of ^{238}U is so large that absorption losses are negligible, and the amount of ^{238}U remains constant.

The resulting model can then be expressed by the following three boxed equations:

$$\frac{dN^{\mathrm{U235}}}{dt} = -N^{\mathrm{U235}}(t)\bar{\sigma}_a^{\mathrm{U235}}\phi$$
$$\longrightarrow \boxed{N^{\mathrm{U235}}(t) = N^{\mathrm{U235}}(0)e^{-\bar{\sigma}_a^{\mathrm{U235}}\phi t}} \qquad (10.27 \rightarrow 10.31)$$

$$\frac{dN^{\mathrm{U238}}}{dt} = -N^{\mathrm{U238}}(t)\bar{\sigma}_a^{\mathrm{U238}}\phi \approx 0$$
$$\longrightarrow \boxed{N^{\mathrm{U238}}(t) = N^{\mathrm{U238}}(0)} \qquad (10.27 \rightarrow 10.34)$$

$$\frac{dN^{\mathrm{Pu239}}}{dt} = N^{\mathrm{U238}}(t)\bar{\sigma}_\gamma^{\mathrm{U238}}\phi - N^{\mathrm{Pu239}}(t)\bar{\sigma}_a^{\mathrm{Pu239}}\phi$$
$$\longrightarrow \boxed{N^{\mathrm{Pu239}}(t) = \frac{\bar{\sigma}_\gamma^{\mathrm{U238}}}{\bar{\sigma}_a^{\mathrm{Pu239}}}N^{\mathrm{U238}}(0)\left[1 - e^{-\bar{\sigma}_a^{\mathrm{Pu239}}\phi t}\right]} \qquad (10.30 \rightarrow 10.37)$$

4 Time, fluence, and burnup

Time is a familiar quantity, but alternatives can be useful when quantifying depletion effects. For example, the **fluence** is the time-integrated flux,

$$\Phi(t) = \int_0^t \phi(t')\, dt', \qquad \text{(FNRP 10.32)}$$

and has units of cm^{-2}. For constant flux, $\Phi(t) = \phi t$.

Burnup is the energy produced per (initial) unit mass of actinides and is listed in a variety of units that include

- MWd/kg (most common),
- GWd/MTU (equivalent to MWd/kg on the same mass basis; MTU is "metric ton (of) uranium"),
- J/kg, and
- % ^{235}U (not burnup, *per se*, but what we use at our TRIGA).

5 Computing burnup with the simplified model

The energy produced per unit volume in one second is the product of the fission-rate density $R_f = \Sigma_f \phi$ [fissions/(cm^{3}s)] and the energy released per fission, $\kappa \approx 3.2 \times 10^{-11}$ J/fission (assuming the usual 200 MeV/fission). For constant flux ϕ, the accumulated energy per unit volume over $t' \in [0, t]$ is

$$Q'''(t) = \int \int_0^t q'''(t')\, dt'$$

$$= \int_0^t \kappa\phi \overbrace{\left(\bar{\sigma}_f^{\mathrm{U235}} N^{\mathrm{U235}}(t') + \bar{\sigma}_f^{\mathrm{U238}} N^{\mathrm{U238}}(t') + \bar{\sigma}_f^{\mathrm{Pu239}} N^{\mathrm{Pu239}}(t')\right)}^{\Sigma_f(t')} dt' ,$$

where $q'''(t)$ is the instantaneous power density.

The initial actinide mass per unit volume is $m_0 \approx \rho_{\mathrm{UO_2}}(238/270)$, though enrichment should generally be considered, and the corresponding burnup is

$$B(t) = \frac{Q'''(t)}{m_0}.$$

6 Reactivity and depletion

Ignoring control and fission products, the multiplication factor can now be written as

$$k(t) = \frac{\left(\nu\bar{\sigma}_{fT}^{\mathrm{U235}} N^{\mathrm{U235}}(t) + \nu\bar{\sigma}_{fT}^{\mathrm{Pu239}} N^{\mathrm{Pu239}}(t)\right)\varepsilon p P_{\mathrm{NL}}}{\bar{\sigma}_{aT}^{\mathrm{U235}} N^{\mathrm{U235}}(t) + \bar{\sigma}_{aT}^{\mathrm{Pu239}} N^{\mathrm{Pu239}}(t) + \Sigma_{aT}^{\mathrm{U238}} + \varsigma \frac{V_m}{V_f} \Sigma_{aT}^{m}}. \quad \text{(FNRP 10.39*)}$$

The T subscripts make the thermal-group cross sections explicit. Make sure to use the appropriate ν for each fissile nuclide.

Important: When doing depletion, we use the total flux over the entire energy range and must use the appropriate *1-group* cross sections. When using the four-factor formula, we need the appropriate *thermal-group* cross sections!

7 Example – Part 1

Consider 4% enriched UO_2 fuel for which the relevant, spectrum-averaged cross sections are

Nuclide	$\bar{\sigma}_a$ [b]	$\bar{\sigma}_f$ [b]
^{235}U	57.20	46.89
^{238}U	1.02	0.11
^{239}Pu	195.10	124.41

Assume the total flux is $\phi = 2 \times 10^{14}$ cm^{-2}s^{-1}. Then

1. determine the number densities of ^{235}U and ^{239}Pu after each of 5 years of operation, and
2. determine the fraction of all power generated due to ^{239}Pu after 1 year of operation.

8 Example – Part 2

Consider the depletion we did above for Example – Part 1. The appropriate *thermal* cross sections for the fuel nuclides are

Nuclide	$\bar{\sigma}_{aT}$ [b]	$\bar{\sigma}_{fT}$ [b]	$\bar{\nu}_T$
^{235}U	297.40	253.09	2.43
^{238}U	1.32	0.00	0.00
^{239}Pu	1146.51	732.22	2.90

Assuming $\varsigma = 1.1$, $V_m/V_f = 1$, $p = 0.6$, $\varepsilon = 1.3$, $P_{\mathrm{NL}} = 0.95$, and $\Sigma_{aT}^m = 0.007$ cm^{-1}, compute

1. the initial reactivity, and
2. the reactivity, fluence, and burnup after each of 5 years of operation.

Ponderable: How does the *fission rate* change in time?